

IMPORTANCE OF SCIENTIFIC FEEDING

Definition: feeding animals on the basis of their **measured nutrient requirements** and the **known nutritive value** of available feeds, rather than by custom or convenience.

Write these 8 justifications: ① **Feed is 60–70% of the total cost** of livestock production — the only input where efficiency directly decides profit. ② Ensures the genetic potential of improved breeds is actually expressed. ③ Prevents both **under-nutrition** (delayed puberty, anoestrus, low yield, low immunity, high neonatal mortality) and **over-nutrition** (fatty liver, dystocia, ketosis, acidosis, laminitis). ④ Avoids **deficiency and toxicity** disorders. ⑤ Permits use of **cheap, locally available and unconventional** feeds without loss of production. ⑥ Improves **feed conversion efficiency** and reduces nutrient excretion (N, P, methane) into the environment. ⑦ Guarantees **product quality** — milk fat, SNF, meat quality, shell strength. ⑧ Makes livestock enterprises **predictable and bankable**.

Indian context line worth adding: our animals live largely on **crop residues (paddy and wheat straw)** with a national deficit of ~11% green fodder, ~23% dry fodder and ~29% concentrate. Scientific feeding here means **strategic supplementation** — urea-treated straw, mineral mixture, bypass nutrients — not imported feeding models.

FEEDING EXPERIMENTS — SIX TYPES, AND WHAT EACH ONE CAN PROVE

EXPERIMENT	WHAT IS MEASURED	ANSWERS THE QUESTION
Digestion trial	Nutrients eaten vs nutrients in faeces	How much of the feed is digestible → DCP, TDN
Balance / Metabolism trial	Intake vs faeces + urine (± milk, gases)	How much is actually retained in the body → N, Ca, P, energy balance
Growth (feeding) trial	Body weight gain, feed intake, FCR	Does the feed perform in a live animal?
Production trial	Milk yield & composition, egg number & mass, wool, work output	Effect on the saleable product
Comparative slaughter	Carcass composition of a baseline group vs the fed group	Energy & protein actually deposited — indirect NE determination without a calorimeter (<i>destructive</i>)
Palatability / cafeteria trial	Free-choice intake when several feeds are offered	Voluntary intake and acceptability

Also name: **in-vitro** (Tilley & Terry two-stage), **in-sacco** (nylon bag), **gas production** (Menke) and **mobile bag** techniques as laboratory substitutes for whole-animal trials — cheap, fast, repeatable, but they cannot measure intake or animal response.

EXPERIMENTAL DESIGNS & THE RULES THAT MAKE A TRIAL VALID

DESIGN	WHEN IT IS USED
CRD (Completely Randomised)	Uniform animals; simplest; unequal group sizes allowed
RBD (Randomised Block)	Animals blocked by weight, age, parity or yield , then randomised within block — the commonest livestock design
Latin Square	Each animal receives every treatment in a different order ; controls both animal and period effects; needs treatments = periods = animals
Switch-over / Reversal / Change-over	Lactating cows — each animal is its own control ; excellent when animals are few and costly
Split-plot / Factorial	Two or more factors tested together (e.g. energy level × protein source) with interaction

Non-negotiable rules: ① **Replication** — never one animal per treatment. ② **Randomisation** — removes bias. ③ **Local control** — uniform housing, management and climate. ④ A genuine **control group**. ⑤ Uniform age, sex, weight, breed and physiological status. ⑥ Adequate **pre-experimental (adaptation) period**. ⑦ Accurate **individual** feeding and recording. ⑧ Statistical analysis before any conclusion is claimed.

Latin square caution: because the same animal receives successive treatments, a **carry-over (residual) effect** can confound results — always insert an adequate wash-out/adaptation between periods.

VOLUNTARY INTAKE — THE FACTOR THAT DECIDES EVERYTHING

Nutrient supply = **digestibility** × **voluntary intake**. A highly digestible feed the animal will not eat is nutritionally useless — say this whenever intake is asked.

Regulation of intake: **Physical / distension control** dominates on roughage diets — rumen fill and rate of passage; correlates negatively with **NDF**. **Chemo-metabolic control** dominates on concentrate diets — VFA (especially propionate) receptors, blood glucose, hepatic oxidation, leptin. **Thermostatic control** — heat increment reduces intake in hot weather (Indian summer, buffalo).

FACTOR	EFFECT ON INTAKE
NDF / lignification of the roughage	↑ NDF ⇒ ↓ intake (rumen fill)
Chaffing, grinding, pelleting	↑ intake (faster passage)
Moisture > 70% / poor silage	↓ intake
Dietary CP below ~7–8%	↓ microbial activity ⇒ ↓ intake sharply
Heat stress, illness, pain, poor water	↓ intake
Palatability, frequency of feeding, TMR	↑ intake

DM intake thumb rules (% of body weight): adult cattle **2.5–3.0** · buffalo **2.5** · high-yielding crossbred cow **3.0–3.5** · sheep & goat **3–4** · growing calf **2.5–3.0**.

CONDUCTING A DIGESTION TRIAL — THE PROTOCOL, IN EXAM ORDER

SELECT ANIMALS
min 3-5, male, uniform

PRELIMINARY PERIOD
7-14 days

COLLECTION PERIOD
5-10 days

ANALYSIS
proximate on feed, residue & faeces

1. **Animals:** minimum **3-5** healthy adults of the same species, breed, age, sex and body weight, dewormed and free from disease. **Males (or castrates) are preferred** — in females urine contaminates the faeces and separation is difficult.

2. **Housing:** individual **metabolism stalls / digestion crates** with separate arrangements for faeces and urine; or **faecal collection harness bags** for grazing/large animals.

3. **Preliminary (adaptation) period — 7-14 days:** purpose is to (i) accustom the animal to the ration, stall and handling, (ii) allow the **rumen microbial population to adapt** to the new feed, and (iii) **clear the digestive tract completely of the previous feed**. Longer for roughage diets and for ruminants than for pigs and poultry.

4. **Collection period — 5-10 days (usually 7):** at a fixed hour daily — **weigh the feed offered and the residue left**; collect **total faeces of 24 h**, weigh, mix thoroughly and take a representative **aliquot of 1/10th (10%)**; do the same for urine.

5. **Preservation:** faeces — add **toluene or a few drops of formalin/conc. H₂SO₄** to stop fermentation and N loss; store in a cool place; dry at **60-70 °C** (low temperature, to avoid losing volatile N). Urine — collect in a bottle containing **20-25 mL of conc. H₂SO₄ or HCl** to hold **pH below 3**, preventing loss of ammonia and bacterial breakdown of urea.

6. **Analysis & calculation:** run proximate analysis on the composite feed, residue and faeces samples, then compute the digestibility coefficient of each nutrient, and from these DCP and TDN.

Digestibility coefficient (%) = [(Nutrient in feed - Nutrient in faeces) ÷ Nutrient in feed] × 100

DCP % = CP % × DC of CP ÷ 100 · TDN % = %DCP + %DCF + %DNFE + (%DEE × 2.25)

Why the value is called "apparent" digestibility: faeces contain not only undigested feed but also **metabolic faecal matter** — shed gut epithelium, digestive enzymes, bile, and dead microbes. Subtracting these gives **true digestibility**, which is always **higher** than apparent digestibility. The difference is largest for **protein** and at **low levels of intake**.

DIGESTION TRIAL VS METABOLISM (BALANCE) TRIAL

POINT	DIGESTION TRIAL	METABOLISM / BALANCE TRIAL
Collected	Feed + faeces only	Feed + faeces + urine (± milk, ± gases)
Measures	Digestibility, DCP, TDN, DE	Retention of a nutrient — N, Ca, P, energy balance
Tells you	How much was absorbed	How much was utilised & stored
Level	DE level	ME and NE level

N balance = N intake - (Faecal N + Urinary N + Milk N)

Positive balance = growth, pregnancy, recovery, wool/milk synthesis. **Negative balance** = starvation, fever, protein deficiency, early-lactation high yielder. **Zero (equilibrium)** = maintenance.

Sources of error in any trial: incomplete faecal collection; contamination with urine, hair, bedding or soil; loss of volatile N during drying; unrepresentative sampling and aliquoting; feed residue not accounted; too short a preliminary period; individual animal variation; and a change in intake during the collection period.

Practical tip that scores: always **record and analyse the residue (orts)** separately — animals selectively refuse the stemmy, high-fibre fraction, so residue composition is not the same as the composition of the offered feed.

MEASUREMENT OF DIGESTIBILITY — DIRECT, INDIRECT & LABORATORY METHODS

METHOD	PRINCIPLE	MERITS	DEMERITS
Direct (total collection)	Weigh everything eaten and every gram of faeces voided	Most accurate ; reference method	Laborious, needs crates, impossible on pasture or in large herds; animals stressed by confinement
Indirect — indicator / marker	An indigestible marker is used as an internal reference; only grab samples of feed and faeces are needed, not total collection	No total collection; usable on grazing animals ; less labour	Marker must be truly indigestible, unabsorbed, uniformly mixed and completely recovered — rarely 100% true
Difference method	Digestibility of a basal ration is measured, then the test ingredient replaces a known part of it; digestibility of the ingredient is obtained by difference	Only way to value a single ingredient (e.g. an oil cake)	Associative effects between ingredients distort the answer
In vitro — Tilley & Terry (1963)	Two stages: 48 h with strained rumen liquor + buffer under CO ₂ , then 48 h with acid pepsin	Fast, cheap, many samples, good correlation with in-vivo	Needs a fistulated donor animal; no intake data
In sacco — nylon bag	Weighed sample in a porous nylon bag suspended in the rumen of a fistulated animal and removed at intervals (0-72 h)	Gives the rate and extent of degradation — the basis of RDP/RUP values	Fistulated animals needed; particle loss through pores; no post-ruminal digestion
Gas production (Menke)	Volume of fermentation gas produced in vitro over time	Kinetics of fermentation; screens for methane	Indirect; gas ≠ nutrient supply

Digestibility % = 100 - 100 × [(% Indicator in feed ÷ % Indicator in faeces) × (% Nutrient in faeces ÷ % Nutrient in feed)]

Indicators — learn both lists. Internal (natural to the feed): lignin, acid insoluble ash (AIA / silica), indigestible ADF or NDF, plant **chromogen**, n-alkanes of plant cuticle wax. **External (fed deliberately):** chromic oxide (Cr₂O₃) — the classic, given as ~10 g twice daily in gelatin capsules or a paper packet; **titanium dioxide (TiO₂)**; polyethylene glycol (PEG, for liquid phase); rare-earth elements (Yb, Dy); barium sulphate.

Properties of an ideal indicator (5): totally **indigestible & unabsorbable** · **no pharmacological effect** on the animal or the microbes · mixes **uniformly** and passes at the same rate as the digesta · **quantitatively recoverable** in faeces · easily and accurately **estimated in the laboratory**.

FACTORS AFFECTING DIGESTIBILITY OF A FEED

FACTOR	EFFECT & THE REASON
1. Composition of the feed — the single most important factor	↑ Crude fibre and especially LIGNIN ⇒ ↓ digestibility. Lignin is indigestible and physically shields cellulose & hemicellulose from microbial enzymes. Every 1% rise in lignin drops DM digestibility appreciably
2. Stage of maturity of the fodder	As the plant matures, cell wall (NDF, ADF, lignin) rises and cell contents fall ⇒ both digestibility and intake fall. Hence "cut at 50% flowering"
3. Species & type of animal	Ruminants > horse > pig > poultry for fibrous feeds. Ruminants digest crude fibre 2–3× better than pigs; poultry practically not at all
4. Level of feeding	Feeding above maintenance speeds the rate of passage ⇒ less time for microbial and enzymatic attack ⇒ digestibility falls by ~2–4% per multiple of maintenance . This is why high yielders on heavy intake digest less efficiently
5. Ratio of nutrients / protein level	Dietary CP below 7–8% starves the rumen microbes of N ⇒ fibre digestion collapses. Adding urea or a protein supplement to straw raises the digestibility of the straw itself
6. Associative effect of feeds	Excess readily fermentable starch lowers rumen pH and inhibits cellulolytic bacteria ⇒ negative associative effect on roughage digestion
7. Feed preparation & processing	Chaffing, soaking, alkali/urea treatment, steaming ⇒ ↑ digestibility. Very fine grinding of roughage ↓ digestibility (rapid passage) though it raises intake
8. Fat level	Free oil above 3–4% of DM coats fibre and poisons cellulolytic bacteria ⇒ fibre digestibility falls
9. Anti-nutritional factors	Tannins bind protein & enzymes; saponins, oxalates and mycotoxins depress digestion
10. Mineral & vitamin adequacy	S, P, Co and trace minerals are needed for microbial growth; deficiency ⇒ ↓ digestion
11. Frequency of feeding & water supply	Frequent small meals stabilise rumen pH ⇒ better fibre digestion; water restriction depresses intake & digestibility
12. Work, age, health & individuality	Heavy work and parasitism reduce digestibility; very young and very old animals digest less; individual variation of 2–5% always exists

FEEDING STANDARDS — HISTORY IN ONE TABLE

Definition: a **feeding standard** is a statement of the quantities of nutrients required daily by an animal of a given species, body weight and level of production, to maintain it in the desired state.

STANDARD	WORKER / YEAR	BASIS — THE IDENTIFYING PHRASE
Hay Equivalents	Thaer, 1810	First-ever feeding standard . All feeds expressed in terms of the value of meadow hay
Total nutrients	Grouven, 1859	Based on the total (not digestible) nutrients of the proximate analysis — its fatal flaw
Digestible nutrients	Wolff, 1864	First to use digestible nutrients; single value per animal, no production allowance
Wolff-Lehmann	Lehmann, 1897	Wolff's tables revised for level of production (e.g. per litre of milk)
Starch Equivalent	Kellner, 1905	Net energy for fat deposition ; 1 kg starch = 248 g fat
Therm / Net Energy	Armsby, 1915	Net energy in therms , from respiration calorimetry
Haecker & Savage	1914–20	Specific standards for dairy cattle — DCP + TDN per litre of milk of stated fat %
Morrison	1936 (<i>Feeds & Feeding</i>)	DCP & TDN given as a range with a built-in margin of safety
NRC	USA, revised periodically	Minimum requirement values; species-wise; no margin of safety
ARC / AFRC	UK	Factorial approach in ME and Metabolisable Protein (MP) — scientifically the most rigorous
Kearl	1982	Requirements specifically for ruminants in developing countries
ICAR	India, 1985 / 1998 / 2013	For Indian breeds under tropical, roughage-based feeding — the standard we actually use

Sequence mnemonic: Thaer → Grouven → Wolff → Lehmann → Kellner → Armsby → Haecker → Morrison → NRC → ARC → Kearl → ICAR. "Total → Digestible → Energy → Factorial → Local."

COMPARATIVE EVALUATION — MERITS & DEMERITS OF THE MAJOR STANDARDS

STANDARD	MERITS	DEMERITS
Morrison	Simple DCP + TDN units familiar to Indian teaching; gives a range (minimum–maximum) with a margin of safety ; covers many species and classes in one book	Empirical , derived from practical American feeding experience rather than from measured requirements; TDN overvalues roughages; not adjusted for tropical breeds or climate
NRC (USA)	Based on experimentally determined requirements ; frequently revised on new research; expressed in modern units (ME, NE _L , MP, individual amino acids and minerals); species-specific committees	Gives minimum requirements with no safety margin — must be adjusted for stress; based on temperate breeds, feeds and management ; assumes high-quality feeds not available in India
ARC / AFRC (UK)	Factorial — separately computes maintenance, growth, pregnancy and lactation and sums them; uses ME and Metabolisable Protein (RDP/RUP) , so it is biologically the most sound; accounts for efficiency of utilisation (k _m , k _p , k _l)	Mathematically complex and not field-friendly; requires feed degradability data most Indian laboratories cannot generate
Kearl	Explicitly derived for tropical, developing-country ruminants on crop residues; practical units	Data base limited; not breed-specific
ICAR (India)	Developed on Indian cattle, buffalo, sheep and goat using Indian feeds; accounts for our roughage-based diets and climate; the standard to quote for Indian conditions	Fewer categories than NRC; needs updating for high-producing crossbreds and modern buffalo genetics

The concluding sentence to write: "No single standard is universally applicable — a **feeding standard is a guide, not a law**. It must always be adjusted for breed, climate, housing, stress, feed quality and individual response, and confirmed by monitoring body condition score, production and health."

BALANCED RATION — DEFINITION & CHARACTERISTICS

Definition (write exactly): A **balanced ration** is one which supplies **all the essential nutrients — energy, protein, minerals, vitamins and water — in the required amount and correct proportion**, in a palatable and digestible form, to a given animal **for a period of 24 hours**, so as to enable it to perform the desired function of maintenance and production efficiently and economically.

Characteristics — the 12-point answer:

① **Palatable** and acceptable — otherwise intake fails. ② Adequate **bulk** to satisfy the animal and keep the rumen functioning (never all-concentrate). ③ Slightly **laxative**, never constipating (include wheat bran, green fodder, molasses). ④ Made of a **variety of ingredients** — improves amino-acid balance and palatability. ⑤ **Economical** and built from **locally available** feeds. ⑥ **Free from moulds, mycotoxins, adulterants and anti-nutritional factors**. ⑦ Supplies nutrients in the **correct proportion — appropriate nutritive ratio** for the animal's stage. ⑧ Suited to the **species and physiological state** (growth, pregnancy, lactation, work). ⑨ Contains adequate **green fodder** for carotene and B-complex precursors. ⑩ At least **one-third of the protein from a leguminous or animal source** in young animals. ⑪ Includes **mineral mixture ~2% and common salt ~1%** of the concentrate. ⑫ **Changed gradually** — never abruptly — over 7–14 days, to protect the rumen microflora.

Standard Indian concentrate mixture (write this if asked to formulate): Maize/grain **30–35%** · Groundnut or mustard cake **25–30%** · Wheat bran / rice polish **25–30%** · Deoiled cake or pulse chuni **10%** · Mineral mixture **2%** · Common salt **1%**. Target: **CP 18–20%, TDN 70%** (BIS Type I cattle feed).

BALANCED RATION VS MAINTENANCE RATION VS PRODUCTION RATION

POINT	MAINTENANCE RATION	PRODUCTION RATION
Purpose	Keeps the animal alive, healthy and at constant weight	Supplied over and above maintenance for milk, meat, wool, egg, work or pregnancy
Body weight	No gain, no loss	Supports a defined output
N balance	Zero (equilibrium)	Positive
Composition	Mainly roughage; wide NR (1:8 or more)	Concentrate-supported; narrow NR (1:4 to 1:6)

Practical Indian thumb rules (state these confidently):

- **Maintenance:** ~**1.5 kg** concentrate mixture for an adult cow and ~**2 kg** for a buffalo, plus roughage *ad libitum*.
- **Milk production:** **1 kg** concentrate per **2.5 L** of cow milk (4% fat) and per **2 L** of buffalo milk (7% fat).
- **Pregnancy allowance:** extra **1–1.75 kg** concentrate in the **last 2–3 months**.
- **Green fodder:** **20–30 kg/day** for an adult cow; dry roughage **4–6 kg**.
- **Roughage : concentrate ratio** — maintenance **80:20**, average production **60:40**, high yielder **50:50** (never below 40% roughage).

Never write a ration without checking three things: total **DM intake** is within the animal's capacity (2.5–3% BW), the **nutritive ratio** suits the physiological stage, and **minerals and salt** have been added. Examiners deduct marks for a ration that is nutritionally correct but physically impossible to eat.

LAST-10-MINUTES DRILL — UNIT 2

Feed share of production cost — **60–70%**
Preliminary period of digestion trial — **7–14 days**
Collection period — **5–10 days (usually 7)**
Minimum animals per trial — **3–5**
Sex preferred in a digestion trial — **Male / castrate**
Faecal aliquot taken — **1/10th (10%)**
Faeces preservative — **Toluene / formalin**
Urine preservative & pH — **Conc. H₂SO₄, pH < 3**
Faeces drying temperature — **60–70 °C**
Why "apparent" digestibility — **Metabolic faecal matter not deducted**
True digestibility is — **Always higher than apparent**
Classic external indicator — **Chromic oxide, 10 g twice daily**
Internal indicators — **Lignin, AIA, chromogen, iADF**
In-vitro two-stage method — **Tilley & Terry, 1963 (48 h + 48 h)**
Technique giving RDP/RUP — **In-sacco nylon bag**
Most important factor in digestibility — **Lignin / crude fibre content**
Effect of ↑ feeding level — **↓ digestibility (faster passage)**
Minimum dietary CP for rumen function — **7–8%**
Fat level that depresses fibre digestion — **Above 3–4% of DM**
First feeding standard ever — **Thaer's Hay Equivalents, 1810**
Standard based on total nutrients — **Grouven, 1859**
First digestible-nutrient standard — **Wolff, 1864**
Standard with a margin of safety — **Morrison**
Standard with NO safety margin — **NRC**
Factorial ME + MP standard — **ARC / AFRC (UK)**
Standard for developing countries — **Kearl, 1982**
Indian standard — **ICAR (2013)**
Design where animal is its own control — **Switch-over / reversal**
Design controlling animal + period — **Latin square**
DM intake of adult cattle — **2.5–3% of body weight**
DM intake of sheep & goat — **3–4% of body weight**
Mineral mixture & salt in concentrate — **2% and 1%**
Concentrate per litre of cow milk — **1 kg per 2.5 L (4% fat)**
Concentrate per litre of buffalo milk — **1 kg per 2 L (7% fat)**
Maintenance concentrate, cow — **~1.5 kg + roughage ad lib**
Roughage:concentrate for high yielder — **50:50 (never <40% roughage)**
Time to change a ration — **Gradually over 7–14 days**
N balance at maintenance — **Zero / equilibrium**
NE method needing no calorimeter — **Comparative slaughter**
Ideal indicator property — **Indigestible, inert, uniformly mixed, fully recoverable, easily assayed**